

LESSON 5.8

SOLVING PROBLEMS INVOLVING DIMENSIONS AND PERIMETERS OF GEOMETRIC FIGURES

LESSON INTRODUCTION

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

LEARNING TARGETS

- I can solve mathematical and real-world problems involving dimensions and perimeters of geometric figures.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

ESSENTIAL QUESTIONS

- How can the scale factor be used to calculate dimensions and perimeter in scale drawings?
- How can the scale factor of a scale drawing be used to solve real-world problems involving perimeter?

LESSON NARRATIVE

This lesson extends students' understanding of scale factor as it applies to dimensions and perimeter when solving real-world problems. The scales used in this lesson include fractional scales, as students build a deeper understanding of how to apply the concepts they have learned to solve problems. Students continue to use equivalent-ratio thinking in solving the problems presented in this lesson collaboratively before practicing on their own.

COMPONENT SUMMARY

5.8.1 WARM-UP (~5 MIN.)

Notice and Wonder using a map overlaid with a grid

5.8.3 COLLABORATION (15-20 MIN.)

Work collaboratively to solve real-world problems applying scale factor to dimensions and perimeter

5.8.5 WRAP-UP (~5 MIN.)

Create ratios to describe two rectangles and determine if there is a common ratio

5.8.2 GUIDED INSTRUCTION (10-15 MIN.)

Determine how to use the scale factor to determine the actual perimeter from a scale drawing

5.8.4 YOUR TURN! (10-15 MIN.)

Calculate perimeters and scale factors

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Grid paper

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may construct ratios incorrectly, for example, reversing the order of the numbers.

THINGS TO CONSIDER

- Consider modeling how to apply knowledge of equivalent ratios to solve the problems in this lesson. Examples are shown throughout the guide.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

The 5.8.1 Warm-Up is a Notice and Wonder activity in which students observe a map of the Jacksonville area with a grid overlaying it. Engage the class in a discussion and highlight responses that are mathematically interesting. For example: I notice there is no scale. I wonder what each gridline represents in feet, miles, etc. Position students to consider how the scale is important and consider how this relates to scaled copies.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

5.8.3 Collaboration is structured to allow students to discuss how the scale factor as a ratio can be used to determine dimensions as well as perimeter. There are also other opportunities to discuss student ideas and understandings as a way of formatively assessing student learning. This Collaboration lends itself well to establishing a culture in which students ask questions of the teacher and their peers and use errors as an opportunity for learning.

DIFFERENTIATE INSTRUCTION

As students work to determine the dimensions of scaled copies based on real-world scenarios, ensure they are using equivalent-ratio thinking. An example is shown for the first question in the 5.8.3 Collaboration:

$$\frac{\text{actual pool}}{\text{drawing}} = \frac{1 \text{ m}}{2 \text{ cm}} \times \frac{15}{15} = \underline{30 \text{ cm}}$$

$$\frac{\text{actual pool}}{\text{drawing}} = \frac{2 \text{ m}}{5 \text{ cm}} \times \frac{6}{6} = \underline{30 \text{ cm}}$$

5.8.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

Some things that students might notice:

- The grid can assist the estimation of distances between cities.
- The cities seem to land within the boxes on the grid.
- There is no scale.

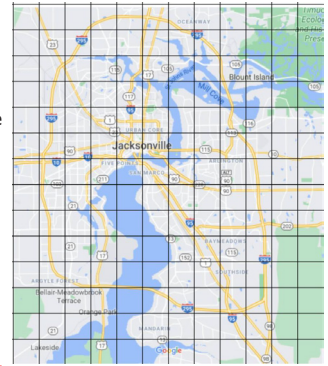
Students might wonder:

- How is the grid helpful in determining distance?
- Why isn't there a scale?



5.8.1 Warm-Up: Notice and Wonder

A map of Jacksonville and its surrounding area is shown.

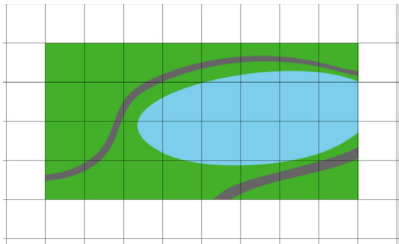


1. What are some features you notice about the map?
Answers will vary. I notice that the roads are yellow.
2. How would the scale help you if you were planning a trip?
Answers will vary. The scale might help me judge how far a distance is and what the most appropriate method for traveling might be – car vs. walking, for instance.
3. How could the gridlines help you determine distance?
The gridlines create a consistent measurement. You could use them to create a scale that would help estimate measurements.



5.8.2 Guided Instruction: Scale Factor as a Ratio

Below is a scale drawing of a park where the scale is 1 centimeter (cm) to 90 meters (m). Each gridline represents 1 cm.



1. Fill in the table with the appropriate dimensions.

	Length	Width	Perimeter
Park Drawing	8 cm	4 cm	24 cm
Actual Park	720 m	360 m	2,160 m

2. Complete the ratio comparing the lengths from the table. Simplify if possible.

$$\frac{720 \text{ m}}{8 \text{ cm}} = \frac{90 \text{ m}}{1 \text{ cm}}$$

3. Complete the ratio comparing the widths from the table. Simplify if possible.

$$\frac{360 \text{ m}}{4 \text{ cm}} = \frac{90 \text{ m}}{1 \text{ cm}}$$

4. Write the perimeters as a ratio. Simplify if possible.

$$\frac{2,160 \text{ m}}{24 \text{ cm}} = \frac{90 \text{ m}}{1 \text{ cm}}$$

5. What do you notice about the three ratios and the scale factor?

Answers will vary. The three ratios all have a scale factor of 90 m : 1 cm.

6. Complete the statement.

The scale or scale factor can be written in the form of a fraction. The ratios of the corresponding sides are ☐ equivalent ☐ not equivalent to the scale factor.

7. Santino tells Sarai that she could have found the perimeter using just the scale factor and the perimeter of the drawing. Explain how Santino could find the perimeter of the actual park using only the scale factor and the perimeter of the drawing.

Santino can find the perimeter of the park by multiplying the perimeter of the scale drawing by the scale factor.
 $(8 + 8 + 4 + 4) \times 90 = 2,160 \text{ m}$

5.8.2 GUIDED INSTRUCTION: SCALE FACTOR AS A RATIO

DIFFERENTIATE INSTRUCTION

Encourage students to draw arrows and label the scale factor they used on the table; this provides an additional visual entry point for learners to keep track of their work and the dimensions they find.

TEACHER MOVES

Some students may need support in answering question 7. Provide further scaffolding through questioning: • How can you use scale factor to find the lengths of sides of rectangles? • Is this similar to how you use scale factor to find perimeter? Why or why not? • If you know the perimeter of the drawing, what can you do with scale factor to find the perimeter of the park?

5.8.3 COLLABORATION: SCALES AND PERIMETER

TEACHER MOVES

Use this section to encourage further discussion from 5.8.2 Guided Instruction question 7. Allow students to discuss multiple ways to determine the dimensions and perimeter.

TEACHER MOVES

Consider pulling a small group, as needed, to model how to use the ratios to determine the new perimeters using strategies learned earlier in the unit. An example is shown:

$$\frac{\text{actual pool}}{\text{drawing}} = \frac{1 \text{ m}}{2 \text{ cm}} \times \frac{15}{15} = \frac{30 \text{ cm}}{15}$$

$$\frac{\text{actual pool}}{\text{drawing}} = \frac{2 \text{ m}}{5 \text{ cm}} \times \frac{6}{6} = \frac{30 \text{ cm}}{6}$$

TEACHER MOVES

When dealing with fractions that become repeating decimals, consider discussing the importance of keeping the value in fraction form to ensure more exact calculations.

TEACHER MOVES

Consider pulling a small group, as needed, to model how to use the ratios to determine the new dimensions using strategies learned earlier in the unit. An example is shown:

$$\frac{\text{actual court}}{\text{drawing}} = \frac{18 \text{ ft}}{3 \text{ in}} \times \frac{2}{2} = \frac{2 \text{ in}}{3}$$

$$\frac{18 \text{ ft}}{3 \text{ in}} \times \frac{5}{9} = \frac{5}{3} \text{ in}$$

$$\frac{\text{actual court}}{\text{drawing}} = \frac{15 \text{ ft}}{1 \text{ in}} \times \frac{2}{2} = \frac{2 \text{ in}}{1}$$

$$\frac{15 \text{ ft}}{1 \text{ in}} \times \frac{5}{3} = \frac{5}{3} \text{ in}$$

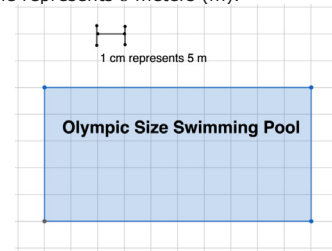
Students could also recognize that the scale factor, 15 ft:1 in will result in a larger actual court and simply verify that its dimensions will fall within the parameters given by the council.



5.8.3 Collaboration: Scales and Perimeter

Work with your partner to complete the following.

1. A scale drawing of an Olympic-size swimming pool is shown on the grid where each 1 centimeter (cm) gridline represents 5 meters (m).

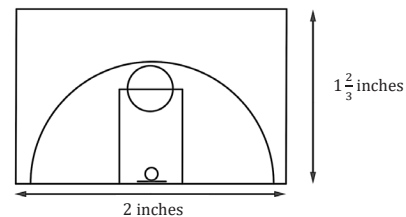


- a. Determine the actual dimensions of an Olympic-size swimming pool in meters (m).
The dimensions of the pool would be 25 m wide and 50 m long.

Shelby wanted to build an Olympic-size swimming pool but did not have the space. So instead, she will build a scaled-down model.

- b. What is the perimeter of the pool in the scale drawing?
The scale drawing of the pool has a perimeter of 30 cm.
- c. What is the perimeter of the actual pool if the scale factor of the drawing was changed to $\frac{1 \text{ m}}{2 \text{ cm}}$?
If the scale factor were adjusted, the scale drawing of the pool would have a perimeter of 15 m.
- d. Calculate the perimeter of the pool if the drawing scale was changed to $\frac{2 \text{ m}}{5 \text{ cm}}$.
If the scale factor were adjusted, the scale drawing of the pool would have a perimeter of 12 m.

2. Vincent proposes an idea to the City Council to install a basketball hoop along with a half-court at his local park. His design is a scale drawing with length of 2 inches (in.) and width of $1\frac{2}{3}$ inches.



- a. What is the actual length and width in feet (ft) of Vincent's planned basketball court if the scale is $\frac{20 \text{ ft}}{1 \text{ in}}$?
Length: 2 inches = 40 feet
Width: $1\frac{2}{3}$ inches = 33 $\frac{1}{3}$ feet
- b. The council tells Vincent his plans will be approved if the court fits on a lot that measures 50 ft long and 25 ft wide. Explain whether Vincent's current plans for the court will fit.
Answers will vary. Vincent's plans will fit within the constraints of the length given by the council but not the width since it will be wider than the allowed 25 feet.
- c. Which scale below will allow Vincent to fall under the parameters given by the council while creating the biggest court? Explain your selection.

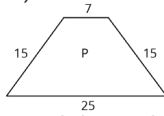
$$\circ \frac{18 \text{ ft}}{3 \text{ in.}}$$

$$\circ \frac{15 \text{ ft}}{1 \text{ in.}}$$

Using the scale $\frac{18 \text{ ft}}{3 \text{ in.}}$, we end with a court of 10 feet by 12 feet. Using the scale $\frac{15 \text{ ft}}{1 \text{ in.}}$, we end with a court of 25 feet by 30 feet. So, Vincent should use the scale $\frac{15 \text{ ft}}{1 \text{ in.}}$.

**5.8.4 Your Turn!**

1. Quadrilateral P is shown, where all measurements are in centimeters (cm).



Quadrilateral R is a scaled copy of Quadrilateral P created using a scale factor of $\frac{5}{2}$. What is the perimeter of Quadrilateral R?

The perimeter of quadrilateral P is 62 cm. The perimeter of quadrilateral R would be that figure multiplied by $\frac{5}{2}$ which totals to 155 cm.

2. Polygon N has a length of 6 units and a width of 2 units.



- a. On the grid, draw a scaled copy of Polygon N using a scale factor of 3.

- b. Explain how the perimeter of the new polygon compares to that of Polygon N.

The perimeter of Polygon N is 16 units. The perimeter of the scaled shape would be 48 units.

5.8.4 YOUR TURN!**DIFFERENTIATE INSTRUCTION**

To provide an additional visual entry point, consider allowing the use of extra grid paper for students to work out and analyze the questions where both figures are neither given nor required to be drawn.

5.8.4 YOUR TURN! (CONTINUED)

TEACHER MOVES

Some students may be able to complete this section without performing calculations by using the dimensions and their knowledge of multiplication facts. Encourage students to show how they are using the scale factor to relate the perimeters as modeled in prior components to show their reasonings.

3. The scale factor and the perimeter of one figure in a relationship between scaled copies are given in the boxes on the left. The boxes on the right contain the corresponding perimeters of the other figure in the scaled relationship.

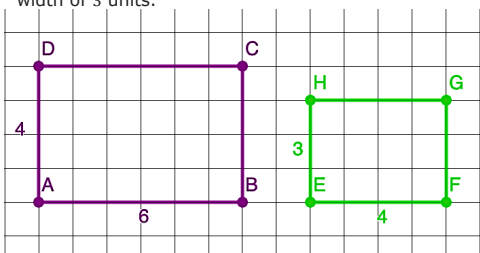
Draw a line connecting each box on the left with the perimeter of the corresponding figure in the relationship on the right.

Given Scale Factor & Perimeter	Corresponding Perimeter
$\frac{1 \text{ cm}}{2 \text{ m}}$ scale, perimeter of 32 m	50 m
$\frac{5 \text{ m}}{2 \text{ cm}}$ scale, perimeter of 20 cm	21 cm
$\frac{1 \text{ cm}}{3 \text{ m}}$ scale, perimeter of 63 m	49 m
$\frac{7 \text{ m}}{3 \text{ cm}}$ scale, perimeter of 21 cm	16 cm



5.8.5 Wrap-Up: Ticket Out the Door

1. Rectangle $ABCD$ has a length of 6 units and a width of 4 units. Rectangle $EFGH$ has a length of 4 units and a width of 3 units.



Kira and Liam are discussing whether they are an example of scale models. Kira says they are, but Liam says they do not fit the definition of scale models.

- a. Complete the ratios.

$$\frac{AB}{EF} = \frac{6}{4}$$

$$\frac{AD}{EH} = \frac{4}{3}$$

$$\frac{\text{perimeter } ABCD}{\text{perimeter } EFGH} = \frac{20}{14}$$

- b. Use the ratios to explain whether Kira or Liam is correct.

Liam is correct. The ratio of the dimensions of the figures would be equivalent to each other and to the ratio of the perimeters of the figures if these were scale models. Therefore, these are not scale drawings. $\frac{6}{4} = \frac{3}{2}$ and $\frac{4}{3} \neq \frac{3}{2}$ and $\frac{20}{14} \neq \frac{3}{2}$

5.8.5 WRAP-UP: TICKET OUT THE DOOR

QUESTIONING

Consider asking students who finish early to write a brief note explaining to Kira why she is incorrect.